

Other Network-Layer Related Protocols

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Address Resolution Protocol (ARP)

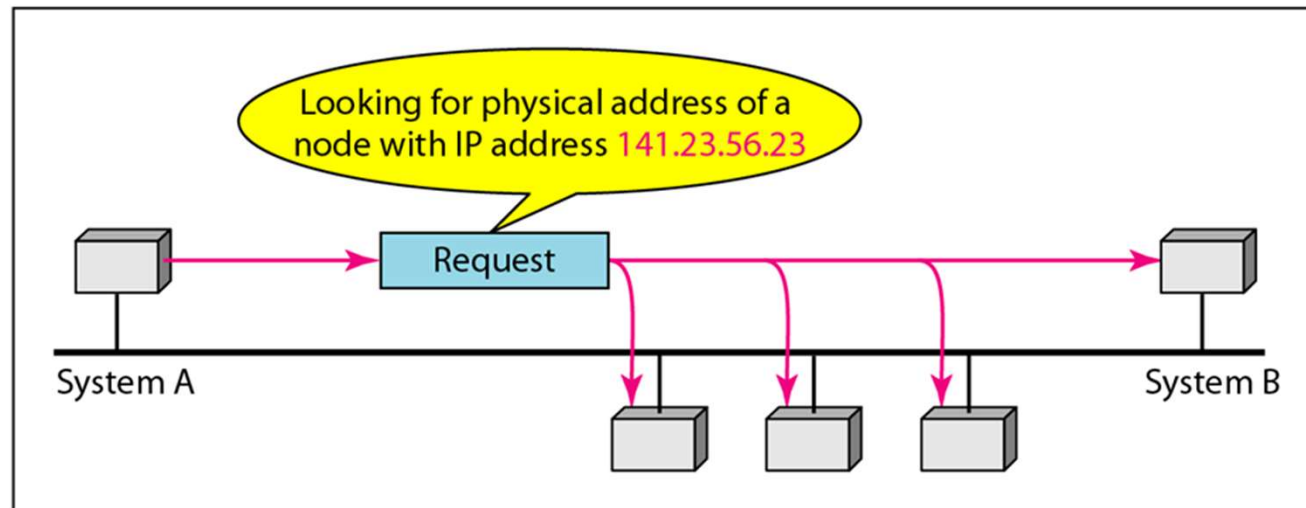
Logical address vs. physical address

- logical address: IP address
 - address used at the Network layer
 - allocated when host is connected to the Internet
- physical address: Ethernet MAC address
 - address used at the Data link layer
 - assigned when the hardware comes out of the factory
- data forwarding at Data link layer
 - physical address is used
 - logical address (IP address) should be mapped to a physical address

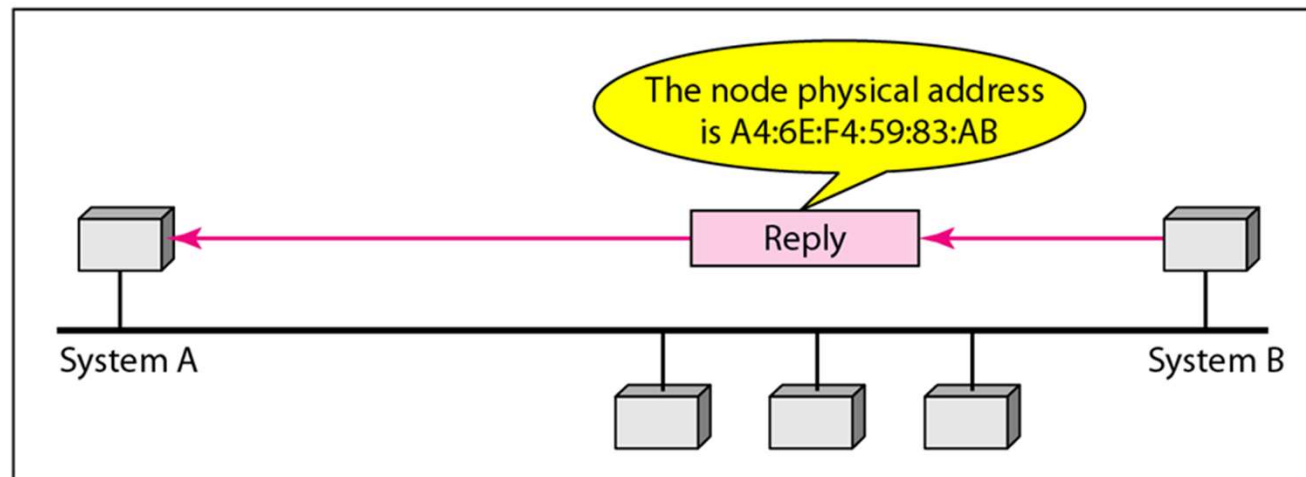
ARP (Address Resolution Protocol)

- Used when the sender knows destination IP address but not the MAC address
- Broadcast ARP request
- The host who has the corresponding physical address sends back ARP reply
- The sender records the physical address in the ARP table
 - So that it does not need to ask every time
- ARP is considered a **Link Layer protocol** (layer 2).

ARP (Address Resolution Protocol)

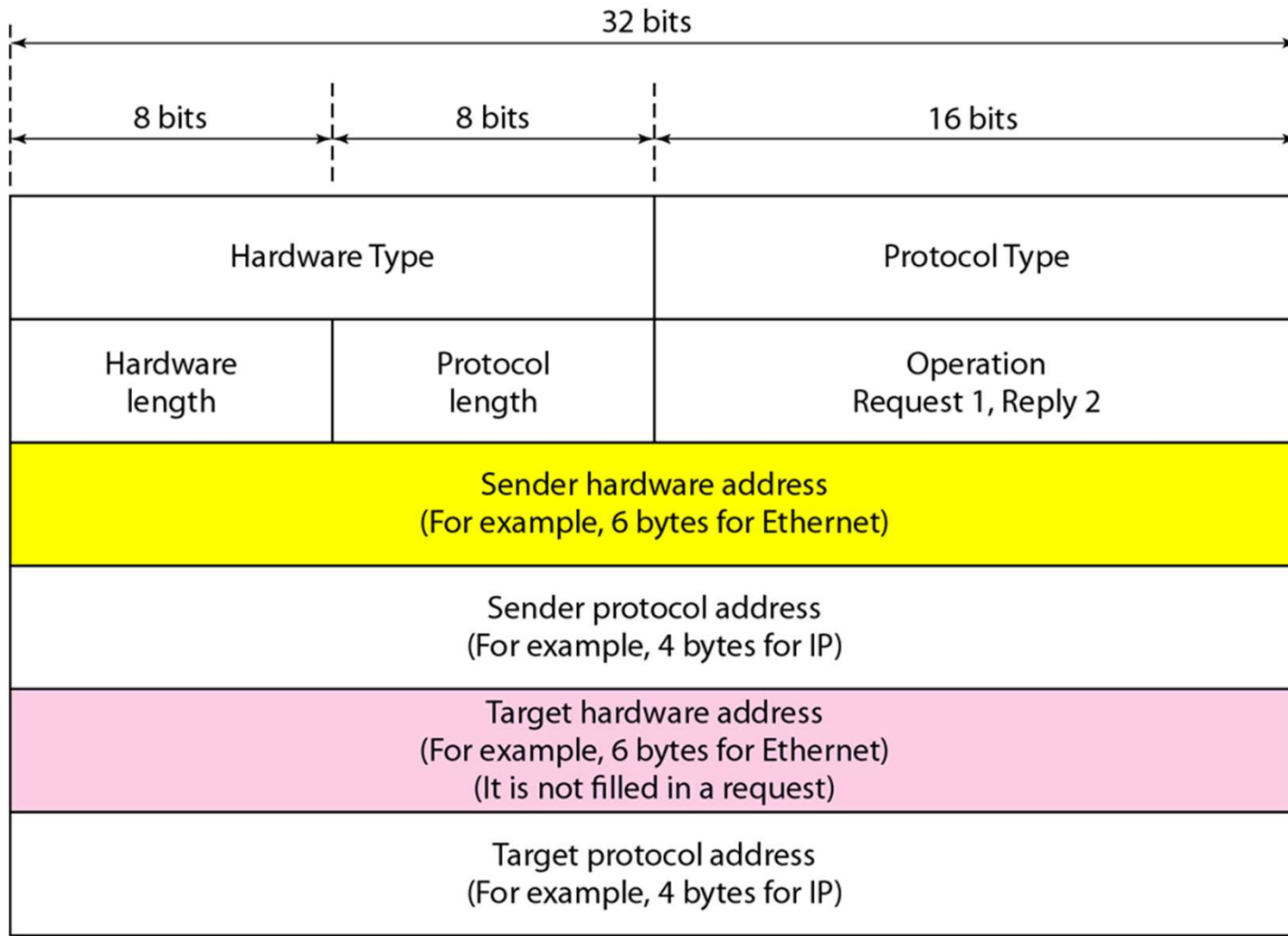


a. ARP request is broadcast



b. ARP reply is unicast

ARP packet format



ARP packet format

- Hardware type: network type
 - e.g.: Ethernet
- Protocol type: Protocol that is using ARP to resolve address
 - e.g.: IP

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

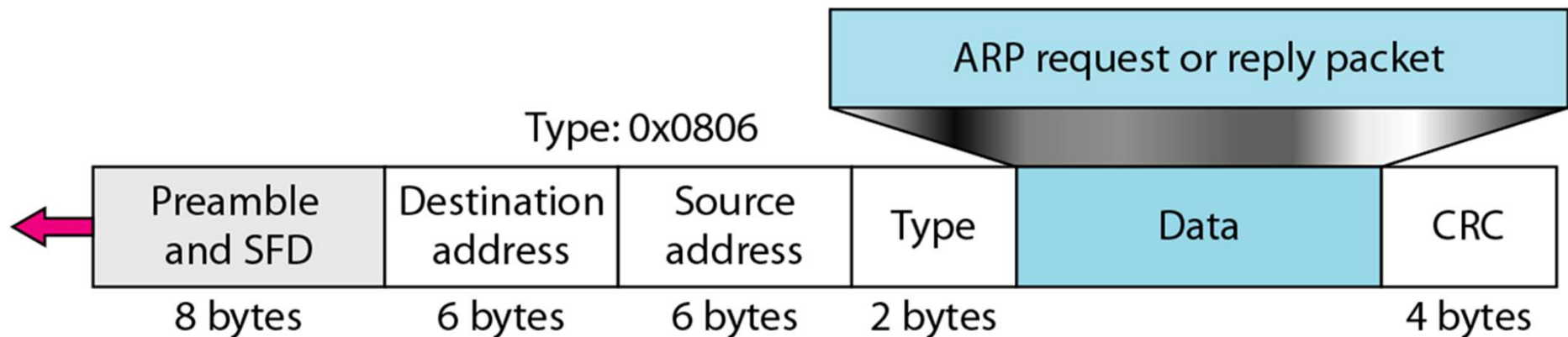
ARP packet format

- Hardware length: physical address length
 - e.g.: Ethernet = 6
- Protocol length: logical address length
 - e.g.: IP = 4
- Operation: ARP request or reply

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

Encapsulation of ARP packet

- ARP packet is encapsulated in the data link frame
- The 'Type' field indicates that the 'Data' is the ARP request/reply



ARP protocol operation

- Source knows Destination's IP address, but not the physical address. (thus cannot send packets.)
- Source generates an ARP request
 - Includes source IP address, physical address, and destination IP address
- ARP request is broadcast
- All hosts in the network receives ARP request
- The host who has the destination IP address replies to the packet. All other hosts discard the packet.

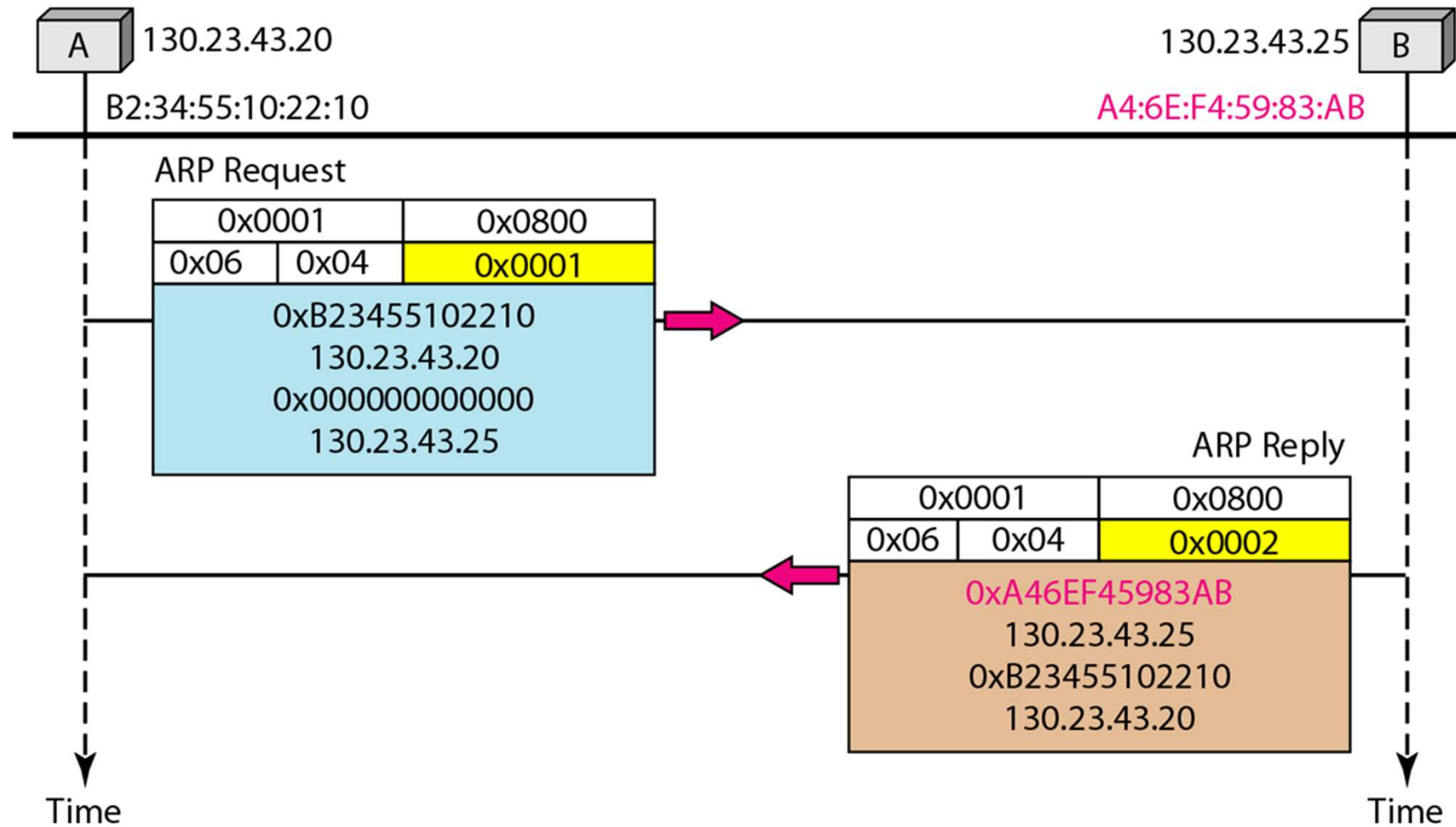
ARP protocol operation

- Destination node sends ARP reply to the Source
 - The packet is sent only to the Source (unicast), since Destination knows Source physical address
- Source finds out Destination physical address when receiving ARP reply

ARP: example

- IP address of a host is 130.23.43.20, and its physical address is B2:34:55:10:22:10.
- This host needs to send to destination whose IP address is 130.23.43.25, and physical address is A4:6E:F4:59:83:AB.
- The source host does not know physical address of the destination host. So it needs to use ARP.
- What will be the contents of ARP request and reply?

ARP: example



Dynamic Host Configuration Protocol (DHCP)

Host Configuration

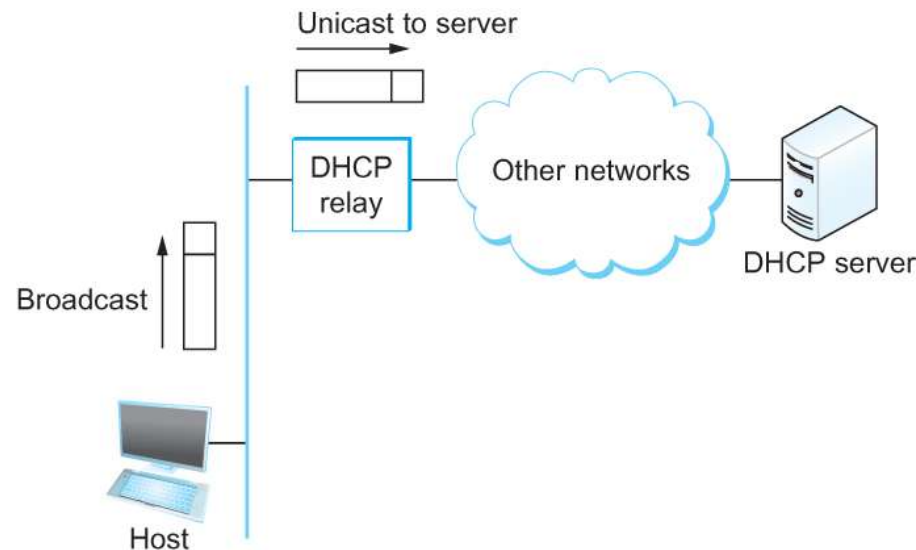
- Ethernet address (physical address)
 - Configured by manufacturer
- IP address (logical address)
 - Must reflect the structure of the internetwork
 - IP address depends on which network the device is currently in
 - When devices moves, IP address must change
 - Manual configuration is burdensome and error-prone
- Automatic configuration of IP address needed

Dynamic Host Configuration Protocol

- DHCP server exists in the network
 - At least one DHCP server for an administrative domain
- DHCP server maintains a pool of available addresses
- DHCP server leases IP address to a host for a certain amount of time
- Hosts may renew the address if they want to continue using the address

Dynamic Host Configuration Protocol

- DHCP is an **Application Layer protocol**
- Newly booted or attached host sends DHCPDISCOVER message to a special IP address (255.255.255.255)
- DHCP relay agent unicasts the message to DHCP server and waits for the response

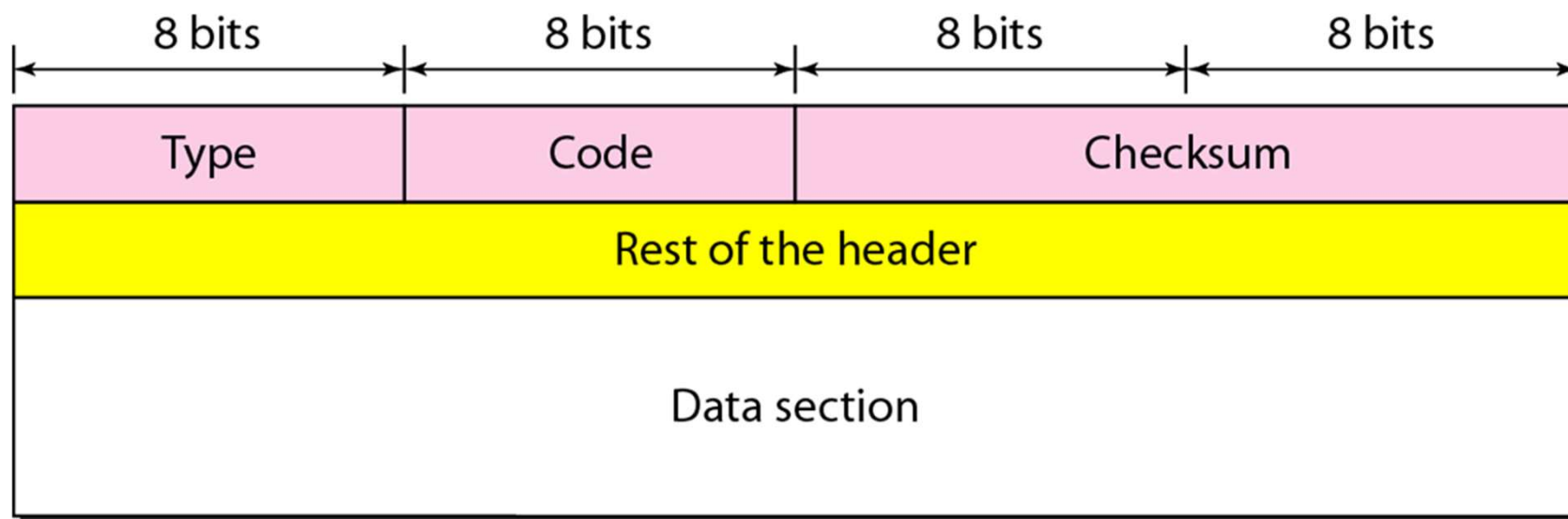


ICMP (Internet Control Message Protocol)

ICMP

- IP does not have a mechanism for reporting errors
 - when the packet is dropped due to error
 - when the packet is dropped due to TTL (Time-To-Live)
 - when the packet is dropped because one of the fragments is lost
- The purpose of ICMP is to report errors to the source
- ICMP is a **Network Layer protocol**

ICMP format

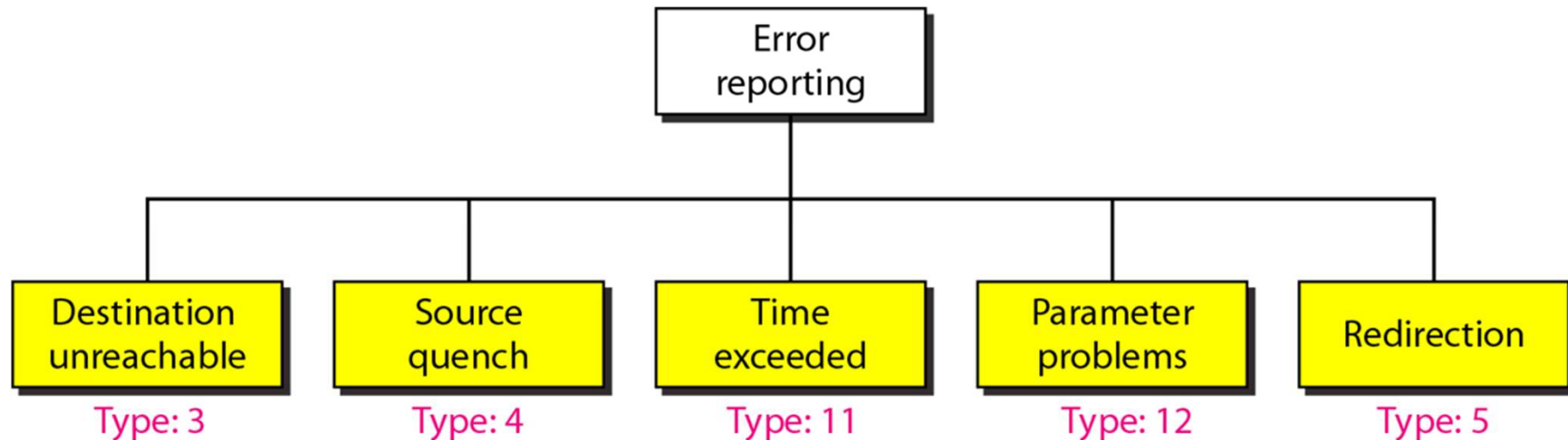


ICMP functions

- Error reporting
- Query

ICMP: error reporting

- ICMP reports to the source IP address under following occasions



ICMP: error reporting

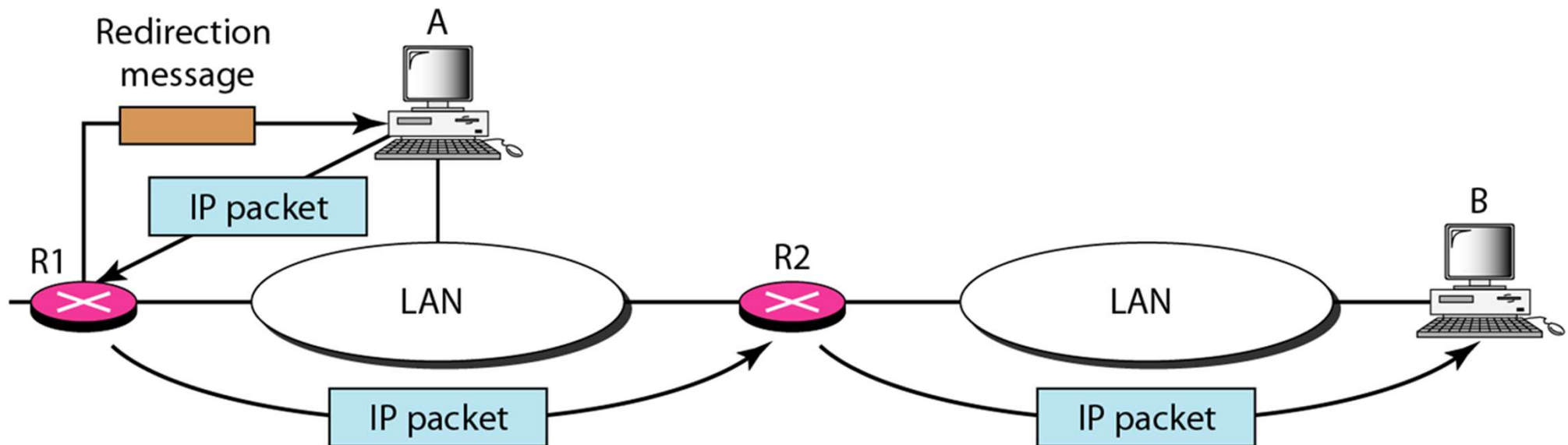
- Destination unreachable
 - Cannot find the next node for the destination
- Source quench
 - Queue of the router or the host is full
- Time exceeded
 - TTL became 0 and the packet is discarded
- Parameter problems
 - Problem in the header fields

ICMP: error reporting

- redirection
 - occurs when there are multiple gateway routers in a network
 - a host often assigns a default gateway without computing the best route
 - if a router finds a better route, it reports to the source node to reassign the gateway router

ICMP: error reporting

- redirection: example
 - Routers R1 and R2 are in the same network with host A
 - Host A's default gateway is R1 (manually assigned)
 - When data is forwarded from A to R1, it finds out that it is better for A to send the packet to R2 for destination host B
 - R1 sends data to R2 and also sends redirection report to A

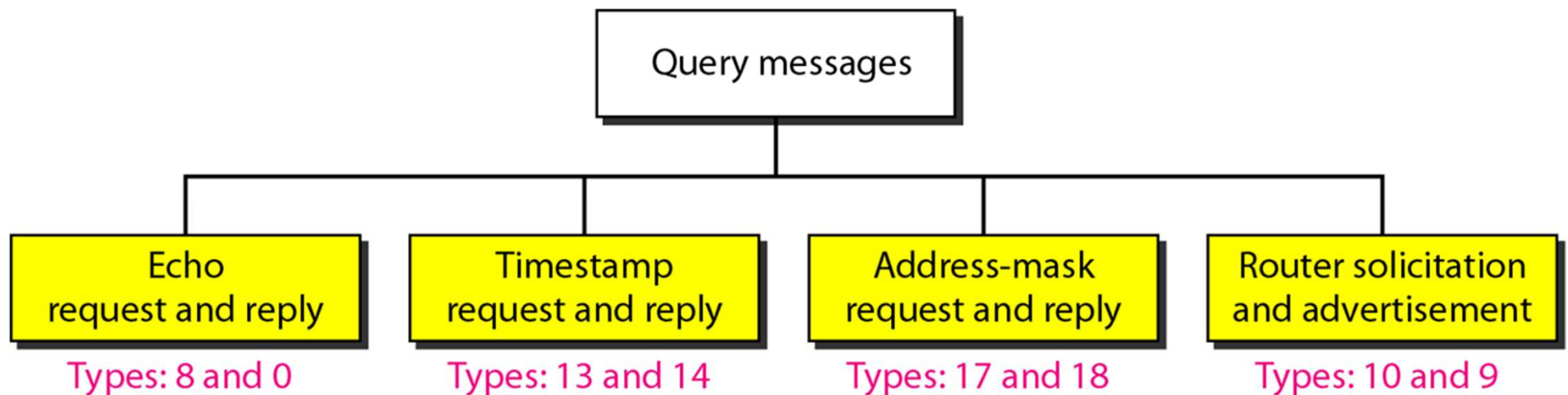


ICMP: error reporting

- ICMP rules
 - ICMP packet is not generated for ICMP packets
 - For fragmented packets, ICMP packet is generated only for the first fragment
 - ICMP is not generated for multicast packets
 - ICMP is not generated for packets with special destination addresses (such as 127.0.0.1 or 0.0.0.0)

ICMP: query

- ICMP can be used to query information



ICMP: query

- Echo request & reply
 - ping
 - used to check if the destination is available
- Time stamp request & reply
 - used to find out round-trip time to the destination

ICMP: query

- Address-mask request and reply
 - When a host knows its IP address, but not the mask
 - It broadcasts address-mask request
 - Any node replies with the address-mask reply
- router solicitation and advertisement
 - A host sends router solicitation to find out what routers are in the network
 - A router sends router advertisement to let the hosts know that the router exists

IP-level debugging tools

- Uses ICMP for debugging purposes
- ping
- traceroute

ping

- An application to send ICMP messages to a destination host to find out if it is available
 - uses ICMP echo request / reply

```
C:\Users>ping www.google.com

Ping www.google.com [74.125.128.147] 32바이트 데이터 사용:
74.125.128.147의 응답: 바이트=32 시간=115ms TTL=45
74.125.128.147의 응답: 바이트=32 시간=85ms TTL=45
74.125.128.147의 응답: 바이트=32 시간=113ms TTL=45
74.125.128.147의 응답: 바이트=32 시간=131ms TTL=45

74.125.128.147에 대한 Ping 통계:
    패킷: 보냄 = 4, 받음 = 4, 손실 = 0 (0% 손실),
    왕복 시간<밀리초>:
        최소 = 85ms, 최대 = 131ms, 평균 = 111ms

C:\Users>
```

traceroute (tracert)

- An application to trace routers between source and destination hosts

```
C:\Users>tracert www.google.com

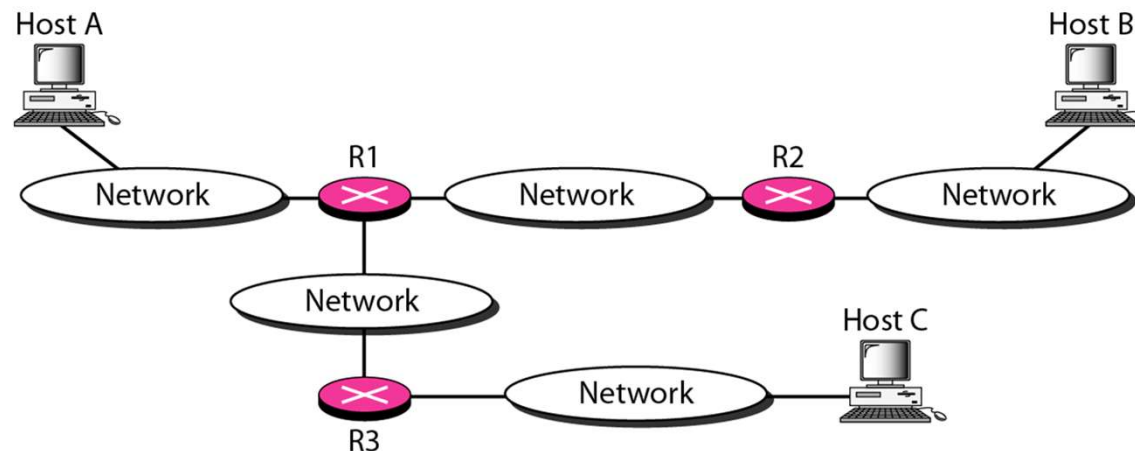
최대 30홉 이상의
www.google.com [74.125.128.104](<으>로 가는 경로 추적:

 1  <1 ms    <1 ms    <1 ms    192.168.0.1
 2   1 ms     1 ms     2 ms     210.115.226.1
 3  <1 ms     1 ms    <1 ms     10.1.1.2
 4   *        *        *        요청 시간이 만료되었습니다.
 5   2 ms     2 ms     1 ms     203.233.37.97
 6   1 ms     1 ms     1 ms     1.208.34.29
 7   5 ms     4 ms     4 ms     1.213.145.97
 8   5 ms     4 ms     4 ms     211.40.6.13
 9   5 ms    14 ms    27 ms     211.63.38.13
10  68 ms    87 ms    82 ms     1.208.105.50
11  65 ms    63 ms    65 ms     72.14.214.101
12  67 ms    62 ms   116 ms     209.85.241.56
13  87 ms    86 ms    83 ms     209.85.253.71
14   *        *        *        요청 시간이 만료되었습니다.
15 134 ms   134 ms   108 ms    hg-in-f104.1e100.net [74.125.128.104]

추적을 완료했습니다.
```

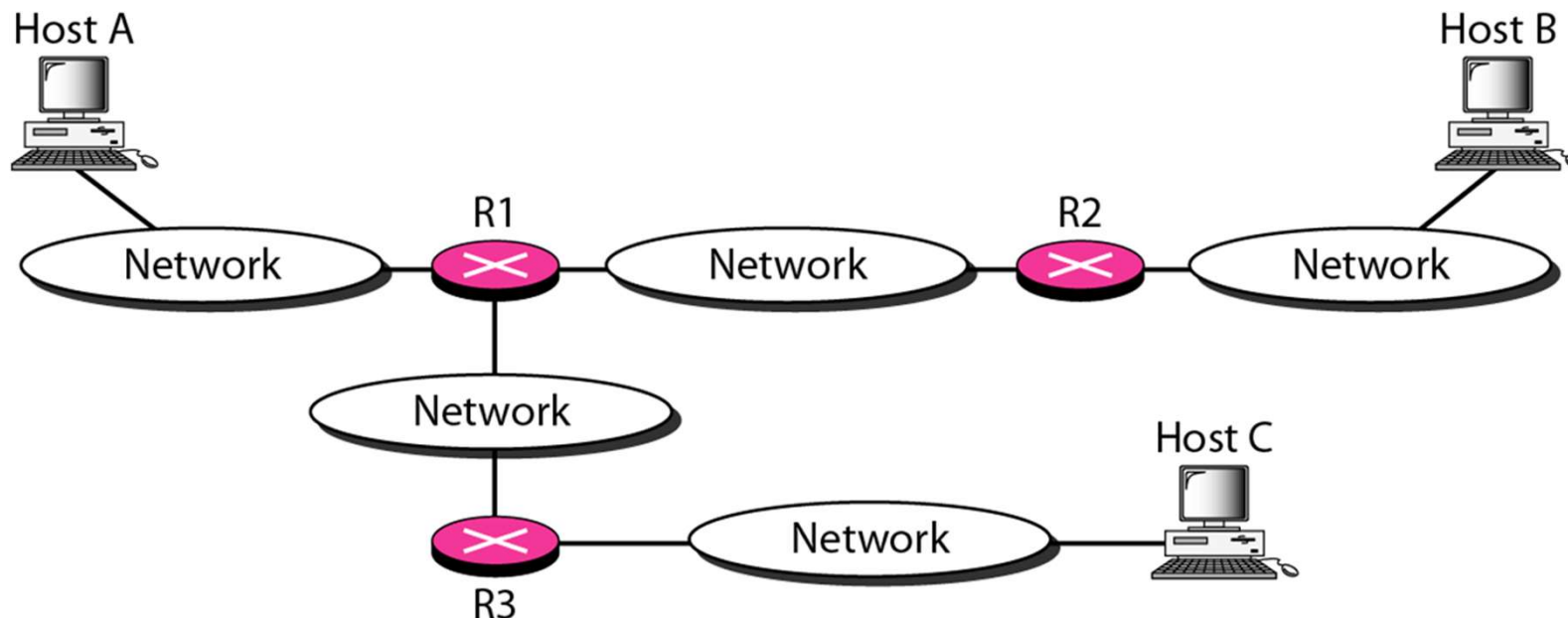

traceroute (tracert) operation

- A sends a UDP packet to B, setting TTL=1.
- The packet is dropped at R1, and an ICMP is sent to A.
 - time exceeded
- Using this ICMP, A knows R1's IP address and the round-trip time from A to R1.
- Multiple UDP packets are sent to get average round-trip time
- When this is done, increment TTL by 1.



traceroute (tracert) operation

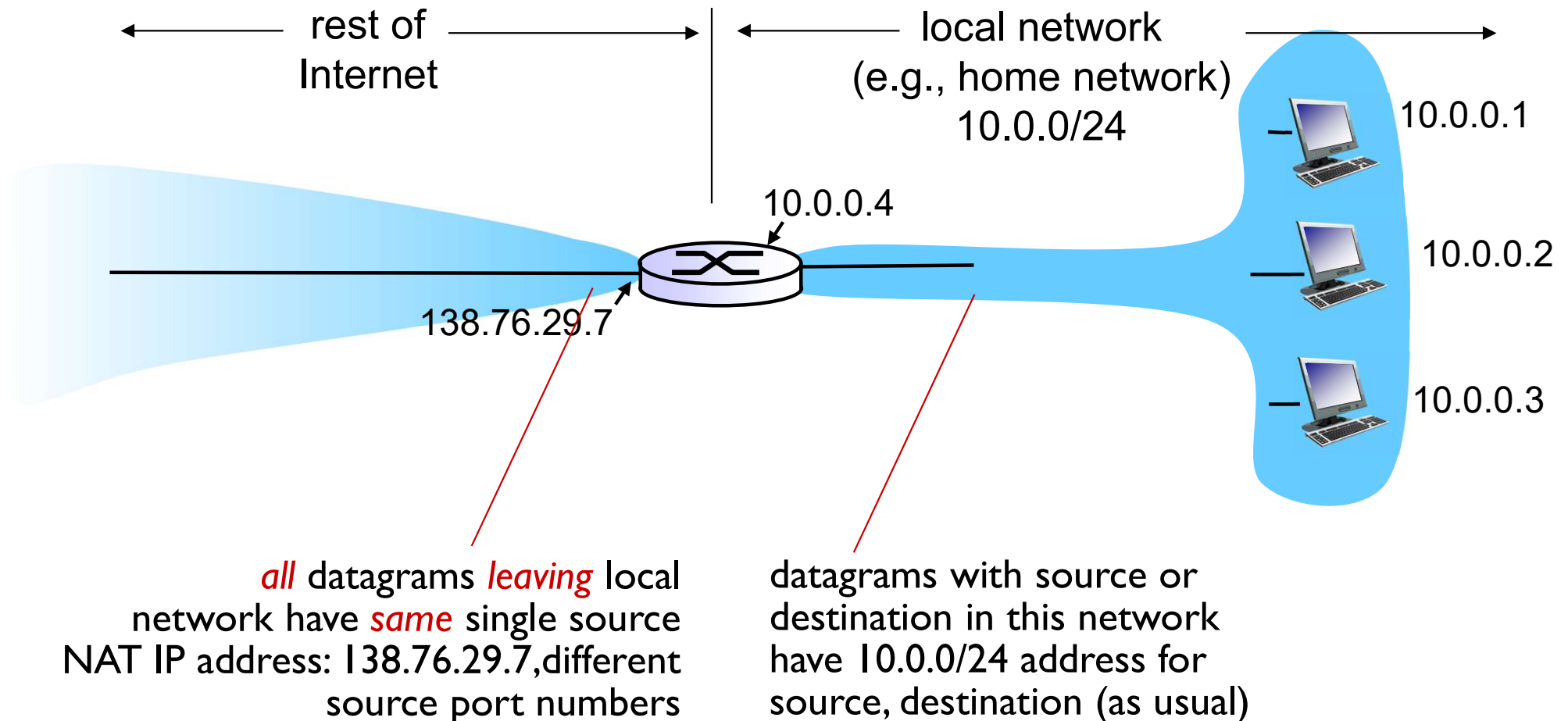
- At the destination host, ICMP (time exceeded) is not send to A.
- However, there is no UDP port receiving traceroute packets at the destination host
- Thus, an ICMP (destination unreachable) is generated and sent to A.



Network Address Translation

NAT: Network Address Translation

- A router can create a local network using network address translation



NAT: Network Address Translation

- Motivation: local network uses just one IP address as far as outside world is concerned
 - range of addresses not needed from ISP: just one IP address for all devices
 - can change addresses of devices in local network without notifying outside world
 - can change ISP without changing addresses of devices in local network
 - devices inside local network not explicitly addressable, visible by outside world (a security benefit)

NAT: Network Address Translation

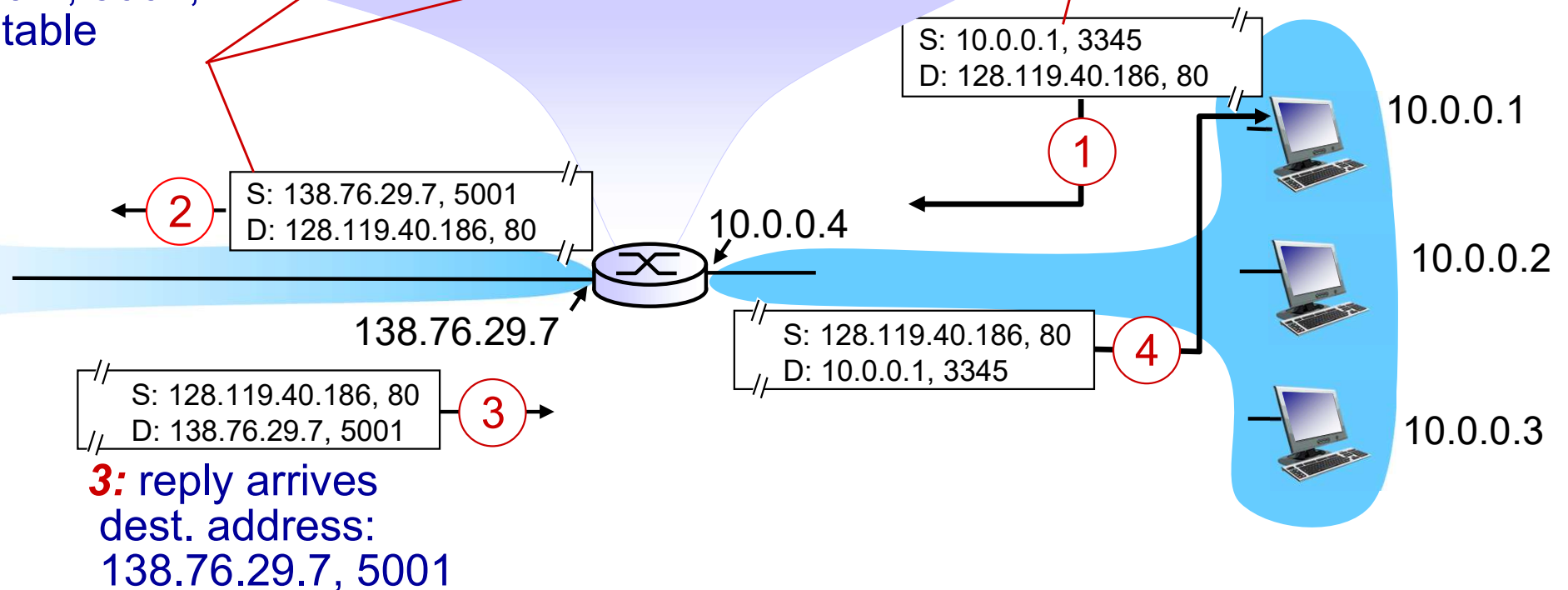
- A NAT router must implement following functions
 - outgoing datagrams: replace (source IP address, port #) of every outgoing datagram to (NAT IP address, new port #)
 - remote clients/servers will respond using (NAT IP address, new port #) as destination addr
 - remember (in NAT translation table) every (source IP address, port #) to (NAT IP address, new port #) translation pair
 - incoming datagrams: replace (NAT IP address, new port #) in dest fields of every incoming datagram with corresponding (source IP address, port #) stored in NAT table

NAT: Network Address Translation

2: NAT router changes datagram source addr from 10.0.0.1, 3345 to 138.76.29.7, 5001, updates table

NAT translation table	
WAN side addr	LAN side addr
138.76.29.7, 5001	10.0.0.1, 3345
.....	

1: host 10.0.0.1 sends datagram to 128.119.40.186, 80



NAT: Network Address Translation

- Private IP address range
 - 10.0.0.0 - 10.255.255.255 (class A)
 - 172.16.0.0 - 172.31.255.255 (class B)
 - 192.168.0.0 - 192.168.255.255 (class C)
 - Used by local networks
 - Not used as public addresses
- Port Forwarding
 - The problem with NAT is when a server is inside a local network
 - The client cannot address the server
 - The NAT router should manually be configured to forward packets with a particular destination port to a specific local IP address